

JOLAPURAM JAYAVARDAN

Associate Software Engineer | Bluetooth Middleware & Controller Firmware | AOSP Stack · HCI · Embedded C/C++
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PROFESSIONAL SUMMARY

Associate Software Engineer at Harman International working across the Bluetooth stack, from AOSP middleware down to controller firmware. Hands-on with HFP, A2DP, AVRCP, MAP and PBAP, Bluetooth Classic and BLE, HCI snoop and Ellisys sniffer analysis, firmware loading via UART, OTA capture, PTS certification, and root cause analysis against the Bluetooth SIG core specification. Strong embedded foundation: bare-metal ARM Cortex-M4, register-level C++ drivers, FreeRTOS and custom bootloaders. ECE graduate, trained at Vector India.

EXPERIENCE

Associate Software Engineer · Harman International, Bengaluru [oct 2026] – Present

- Develop and maintain Bluetooth middleware on Android AOSP (HFP, A2DP, AVRCP, MAP, PBAP) in Java and C/C++, covering feature development and profile bug fixes
- Support Bluetooth controller and firmware work: firmware loading via UART, HCI snoop and Ellisys capture analysis, and signal verification with a logic analyzer
- Perform root cause analysis on field and test issues using the Bluetooth SIG core specification, OTA logs, ADB and Perfetto
- Run PTS (Profile Tuning Suite) certification cycles for Classic and BLE profiles

Bluetooth Developer Intern · Harman International, Bengaluru Jan 2026 – [oct 2026]

- Implemented and debugged HFP, A2DP, MAP, PBAP and AVRCP on the Android Bluetooth stack
- Captured and analysed OTA logs, supported PTS testing, and debugged in Java using ADB

Embedded Software Trainee · Vector India, Bengaluru Completed

- Interrupt-driven firmware and peripheral drivers for SPI, I2C, UART, CAN on ARM7 LPC2129
- FreeRTOS task design, TCP/IP networking fundamentals, and Linux system programming
- Strong grounding in memory management, interrupts, and peripheral interfacing

TRAINING

ARM Cortex-M4 Bare Metal Programming · Argyan Tech Completed

- Cortex-M4 architecture deep-dive: registers, NVIC, exceptions, AAPCS, startup code, linker scripts
- Built bare-metal task scheduler using SysTick & PendSV; debugged with GDB/OpenOCD

TECHNICAL PROJECTS

STM32F407 Bare-Metal C++ Driver Library 2026

Embedded C++ · STM32F407VG · Register Programming · PlantUML

- From-scratch, register-level peripheral drivers (GPIO, RCC, EXTI, NVIC, SysTick, USART, SPI, I2C) with no HAL or LL dependency
- Object-oriented C++ API built directly against the reference manual; documented with PlantUML diagrams, theory notes and working examples

Custom Bootloader Development — STM32F407VG 2026

Embedded C · UART · ARM Cortex-M4 · Python

- UART-based bootloader: flash erase, memory write, CRC verification and application jump; custom VTOR relocation, linker scripts and startup code
- Python host utility for firmware transfer and bootloader command handling over UART

Bare-Metal Task Scheduler — STM32F407VG 2026

GCC ARM CLI · OpenOCD · GDB

- Built with no HAL/IDE: custom linker scripts and startup files; round-robin scheduling via SysTick and manual Cortex-M4 stack frame manipulation
- Flashed and debugged via CLI with live register and stack frame inspection using GDB + OpenOCD

TECHNICAL SKILLS

Languages: Embedded C, Embedded C++, C++, Java, Python

Microcontrollers: STM32F407VG, ARM7 LPC2129

Protocols: SPI, I2C, UART, CAN, Bluetooth Classic & BLE (HFP, A2DP, AVRCP, MAP, PBAP, HCI, SIG Core Spec, OTA, PTS, AOSP Stack, Controller, Firmware Loading via UART)

Tools: ADB, Perfetto, Ellisys Analyzer, HCI Snooper, Logic Analyzer, GCC ARM, OpenOCD, GDB, SEGGER SystemView, Git

Concepts: Middleware, Controller Firmware, Embedded Systems, Device Drivers, Bare Metal, RTOS (FreeRTOS), Interrupts, Memory Management, Linker Scripts, Custom Bootloaders, Linux System Programming

Software Practices: CI/CD Pipelines, Jira, Agile / SDLC, Git & GitHub, Debugging, Root Cause Analysis (RCA), Bug Tracking

CERTIFICATIONS & ACHIEVEMENTS

- Embedded Systems — Eduskills & Microchip; ARM Cortex, STM32 MCU1 & MCU2 — FastBit; Git & GitHub — GeeksforGeeks
- 2nd Place — Circuit Debugging, Ripple 2K24

EDUCATION

B.Tech — Electronics & Communication Engineering — RGM CET, Nandyal CGPA: 8.0/10	<i>2022 – 2025</i>
Diploma — Electronics & Communication Engineering — Sri Jyothi Polytechnic, Vuyyuru 89.2%	<i>2019 – 2022</i>